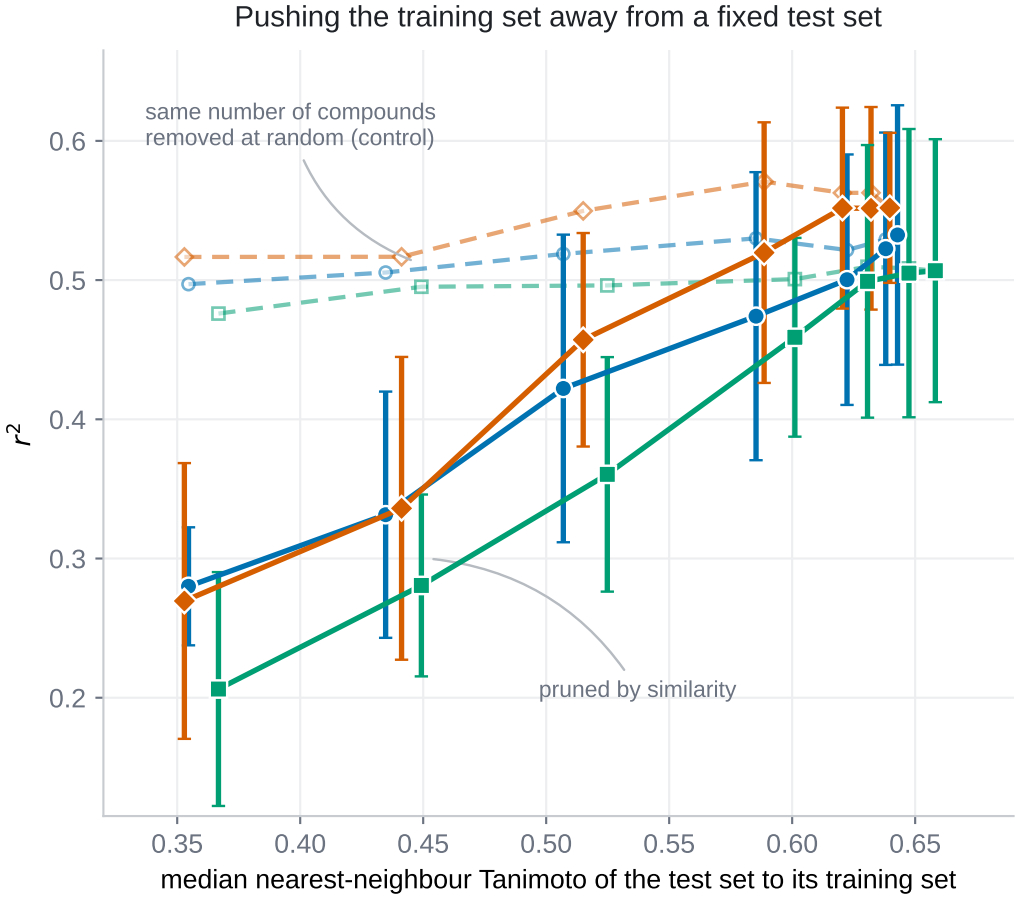
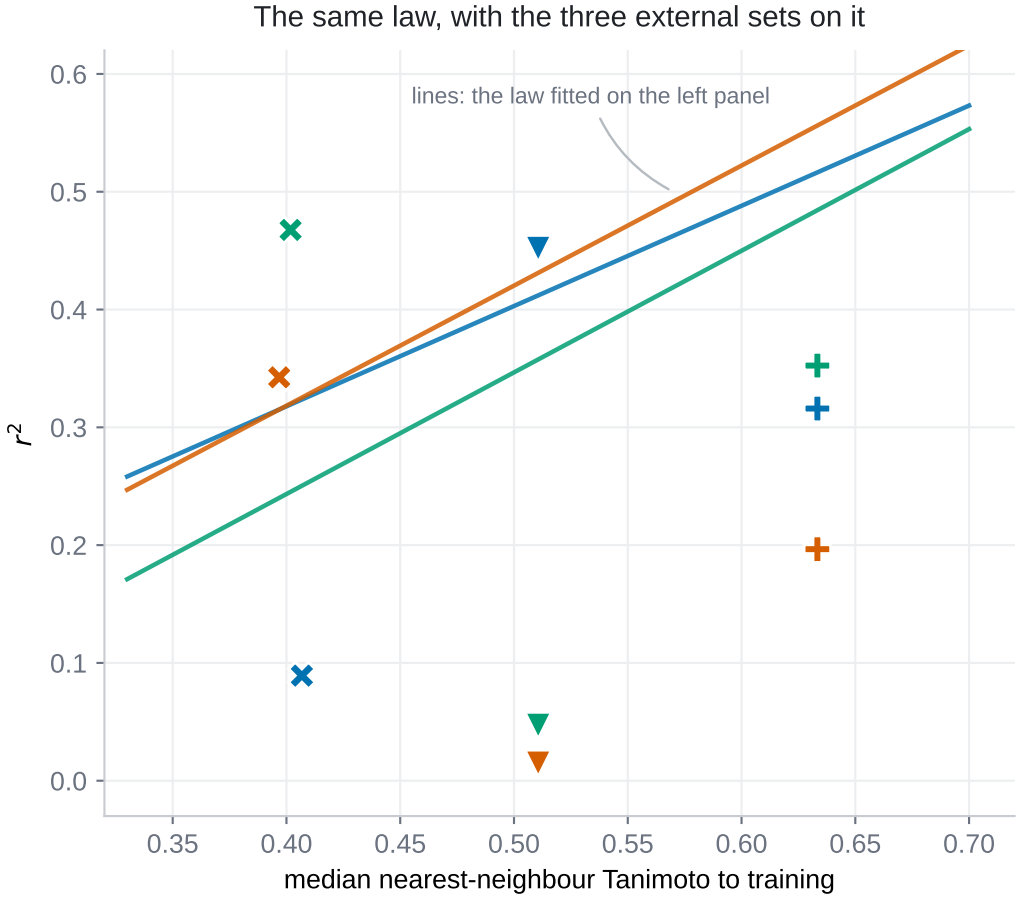
Distance to training predicts accuracy — until the data is really external

Left: a scaffold-grouped test set is drawn, then every training compound within Tanimoto > τ of any test compound is removed, for τ from 1.00 down to 0.40; six replicates, error bars 1 SD. Each 0.10 of lost similarity costs 0.085–0.103 of r^2 (r = 0.74–0.81, p < 1e-7). The dashed control removes the same number of compounds at random and stays flat, so the effect is distance, not training-set size.

Right: across the boundary into genuinely external data the law breaks — r^2 against nearest-neighbour Tanimoto is r = −0.03, while against response-range coverage it is r = +0.55. Inside the training distribution similarity is what limits the model; outside it, coverage is.





DAT NET SERT Stalberga 2026 PDSP / BindingDB Cui et al. 2026